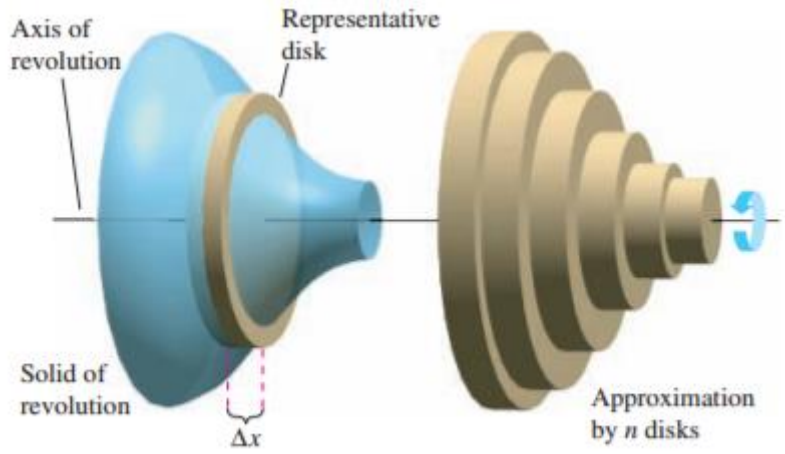
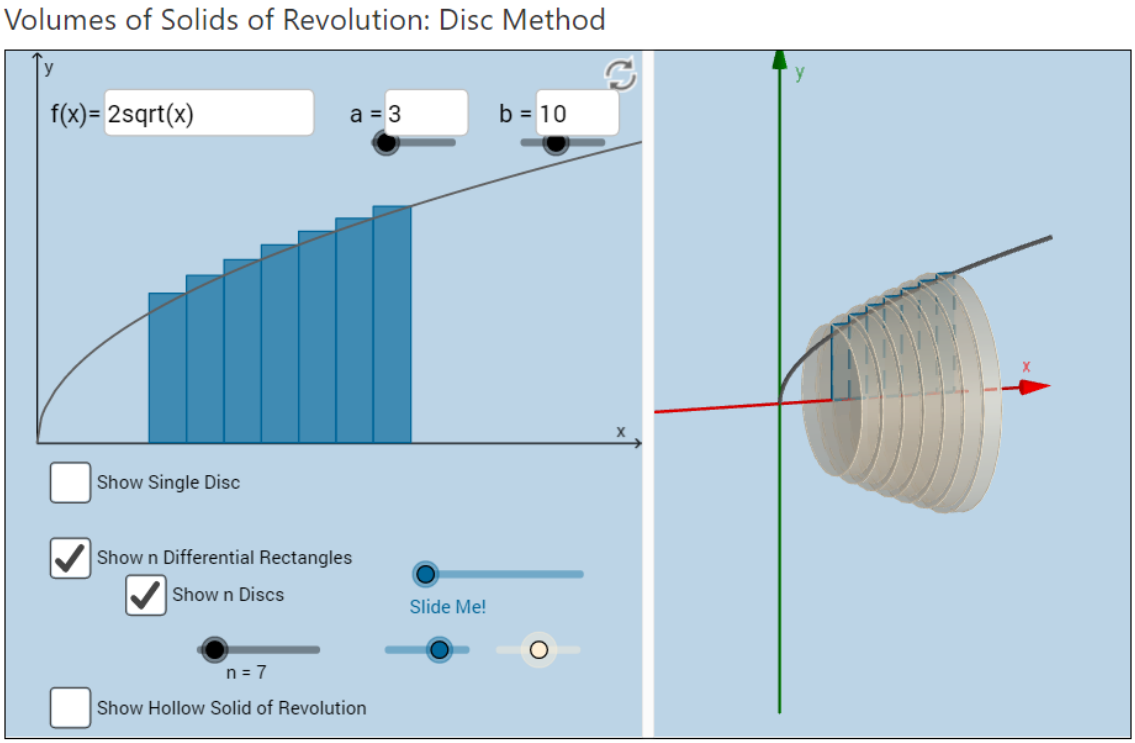
Volumes of Revolution (旋轉體的體積)

Definite Integration can be used not only to find the area under a curve, but also to find the volume of Solid of Revolution when the curve is rotated along the x-axis (or y-axis):



Demonstration:

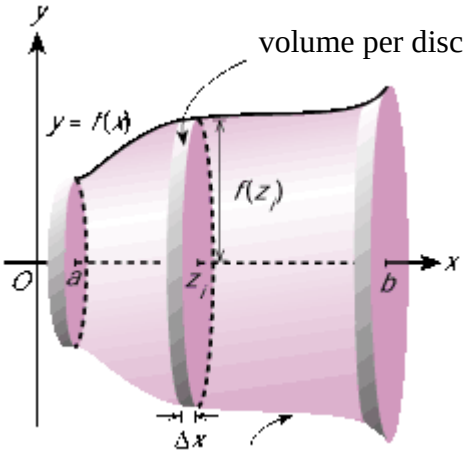
https://www.geogebra.org/m/hhRJQyz9



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Volumes of Revolution

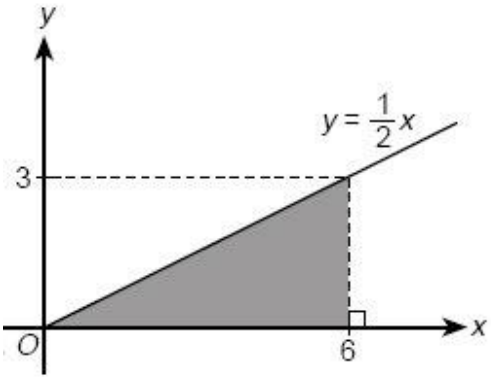
1. Volume of solid of revolution about the *x*-axis



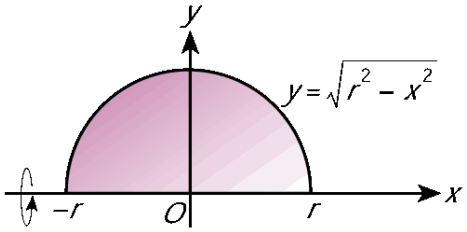
volume per disc = $\pi [f(x_i)]^2 \cdot \Delta x$

$$\text{Volume} = \pi \int_a^b [f(x)]^2 \, dx$$

e.g. 1 Find the volume of solid of revolution of the shaded region about the *x*-axis.



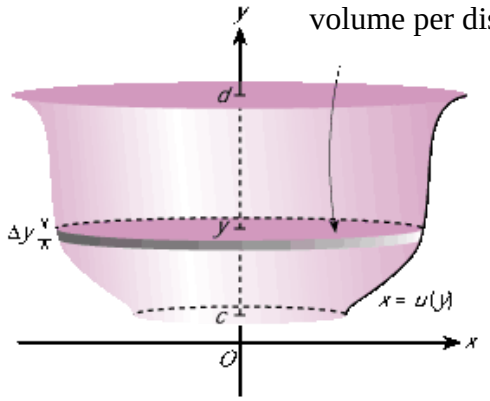
e.g. 2 Find the volume of solid of revolution of the shaded region about the *x*-axis in terms of *r*.



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Volumes of Revolution

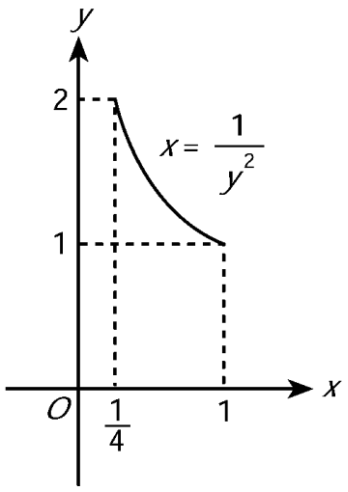
2. Volume of solid of revolution about the *y*-axis



volume per disc = $\pi [u(y_i)]^2 \cdot \Delta y$

$$\text{Volume} = \pi \int_c^d [u(y)]^2 \, dy$$

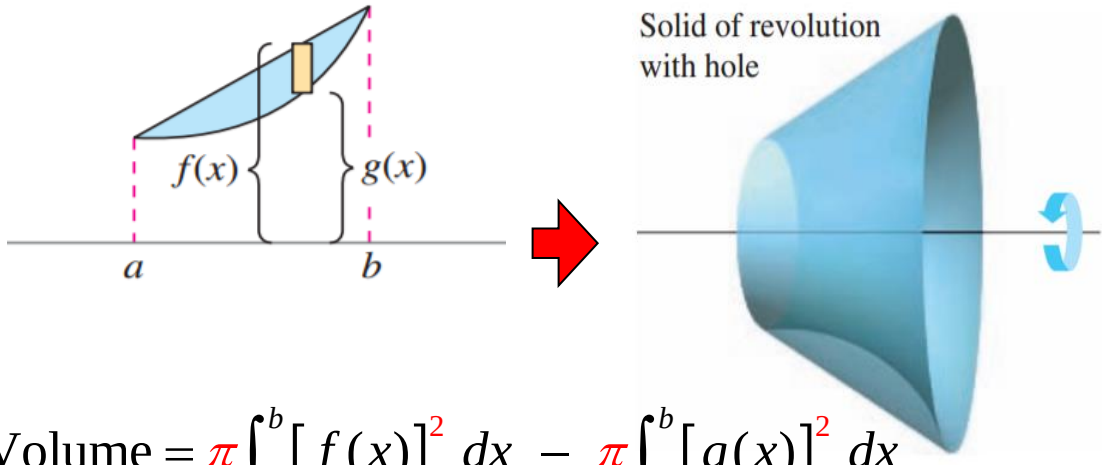
e.g.3 Find the volume of solid of revolution of the area bounded by the curve $x = \dfrac{1}{y^2}$ and the line $y = 1$ and $y = 2$ about the y-axis.



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Volumes of Revolution

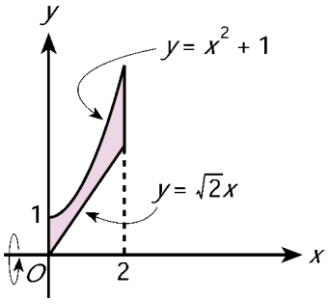
3. Volume of solid of revolution with hole



$$\text{Volume} = \pi\int_a^b [f(x)]^2\,dx - \pi\int_a^b [g(x)]^2\,dx$$

$$= \pi\int_a^b \left\{[f(x)]^2 - [g(x)]^2\right\}\,dx$$

e.g 4 Find the volume of solid of revolution of the shaded region bounded by the curve $y = x^2 + 1$ and $y = \sqrt{2}x^2$ from $x = 0$ to $x = 2$ about the x-axis.

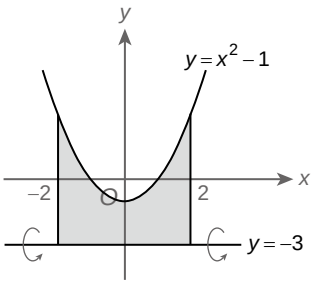


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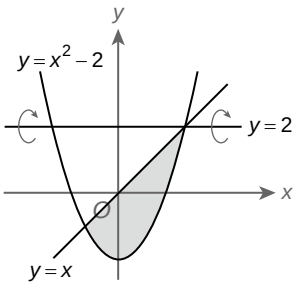
Volumes of Revolution

Extra Examples:

e.g 5 If the shaded region bounded by the curve $y = x^2 - 1$, the lines $x = -2$, $x = 2$ and $y = -3$ is revolved about the line $y = -3$, find the volume of the solid of revolution.



e.g 6 If the shaded region bounded by the curve $y = x^2 - 2$ and the line $y = x$ is revolved about the line $y = 2$, find the volume of the solid of revolution.



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HKDSE Questions (level 1)

1. ***[M2 2012 Q.9]***

(a) Using integration by parts, find $\int x \sin x \, dx$.

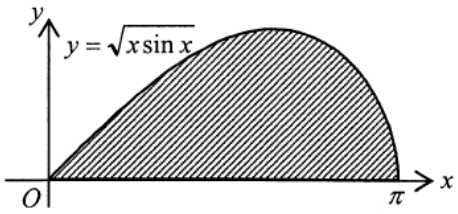


Figure 4

(b) Figure 4 shows the shaded region bounded by the curve $y = \sqrt{x \sin x}$ for $0 \le x \le \pi$ and the x-axis. Find the volume of the solid generated by revolving the region about the x-axis.

(4 marks)

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HKDSE Questions (Level 1)

2. ***[M2 2013 Q.6]***

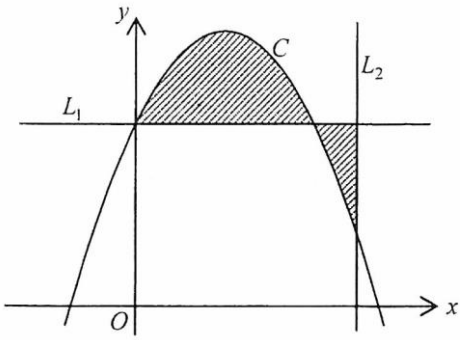


Figure 1

Figure 1 shows the shaded region with boundaries $C : y = \frac{-x^2}{2} + 2x + 4$, $L_1 : y = 4$ and $L_2 : x = 5$. It is given that C intersects L_1 at (0 , 4) and (4 , 4) .

(a) Find the area of the shaded region.

(b) Find the volume of the solid of revolution when the shaded region is revolved about L_1 .

(6 marks)

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HKCEE Questions (Level 1)

3. ***[A.Math 2005 Q.16]***

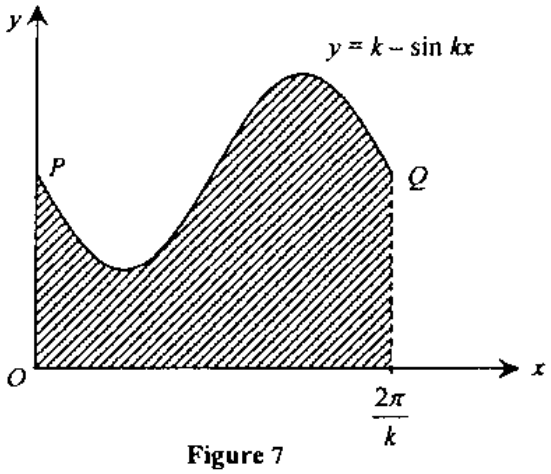


Figure 7

Figure 7 shows the curve $y = k - \sin kx$ for $0 \le x \le \frac{2\pi}{k}$, where $k>1$. P and Q are the end points of the curve.

(a) (i) Find the coordinates of points P and Q.

(ii) Without using integration, find the area of the shaded region us shown in Figure 7. ***(4 marks)***

(b) A solid is formed by revolving the shaded region in the Figure 7 about the x-axis Let V be the volume of the solid.

(i) Show that $V = \pi^2(2k + \frac{1}{k})$.

(ii) Suppose that $2 \le k \le 3$. Find the greatest value of V as k varies.

(8 marks)

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HKCEE Questions (level 1)

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HKCEE Questions (Level 1)

4. ***[A.Math 1993 Paper II Q.12]***

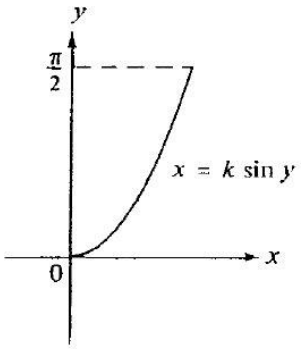


Figure 4 (a)

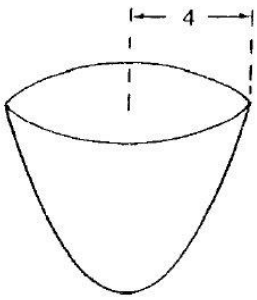


Figure 4 (b)

Figure 4(a) shows the curve $x = k\sin y$, where $k > 0$ and $0 \le y \le \frac{\pi}{2}$. A bowl is formed by revolving the curve about the *y*-axis.

(a) Show that the capacity of the bowl is $\frac{1}{4}k^2\pi^2$ cubic units. ***(6 marks)***

(b) Given that the radius of the rim of the bowl is 4 units. (See Figure 4(b)). The bowl is full of water.

(i) Find the volume of water.

(ii) The water is now pumped out of the bowl at a rate of $(\pi + 2t)$ cubic units per minute, where *t* is the time in minutes after pumping starts.

Find the time taken to pump out half of the water and the time taken to pump out the remaining water in the bowl. Give both answers in terms of π.

(10 marks)

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HKCEE Questions (level 1)

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HKCEE Questions (level 2)

5. ***[A.Math 1991 Paper II Q.11]***

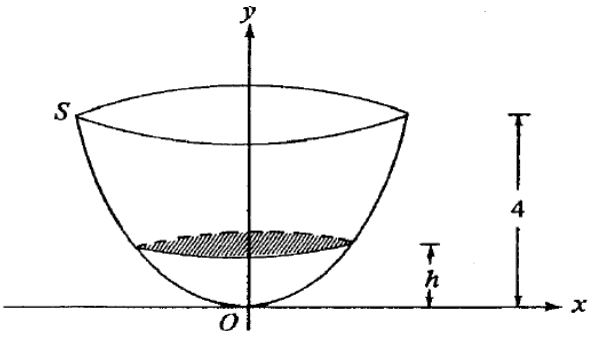


Figure 3(a)

An object *S* is in the shape of the solid of revolution of the region bounded by the curve $x^2 = 4y$ and the line y = 4 revolved about the y-axis, as shown in Figure 3(a).

(a) Find the volume of *S*. ***(3 marks)***

(b) It is given that if *S* is cut by a plane parallel to its top surface at a distance h from O (see Figure 3(a)), the mass of the part of *S* below the plane is given by

$$\int_0^h \pi(16y - 3y^2)dy, \quad \text{where } 0 < h \le 4.$$

(i) Find the mass of *S*.

(ii) If the plane mentioned above cuts S into two parts pf equal volumes, find

(1) the value of *h*,

(2) the ratio of the mass of the lower part to that of the upper part.

(9 marks)

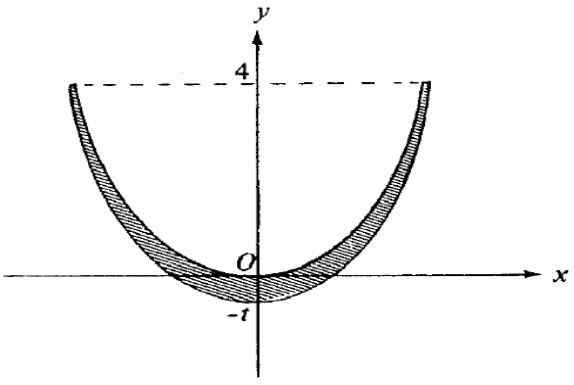


Figure 3(b)

(c) In Figure 3(b), the shaded region is bounded by the curves $x^2 = 4y$, $x^2 = 4(y + t)$ and the line *y* = 4, where *t* is a small positive number. The solid of revolution of the shaded area revolved about the y-axis represents a layer a paint coated on the curved surface of *S*. Show that the volume of paint is approximately equal to $16\pi t$.

(4 marks)

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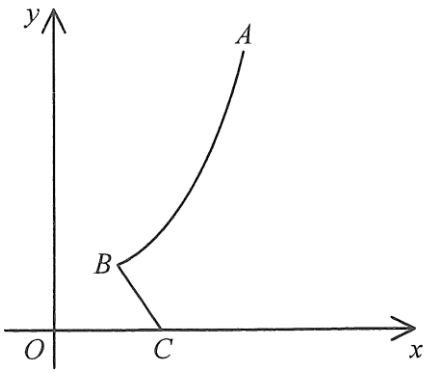
HKCEE Questions (level 2)

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HKDSE Questions (Level 2)

6. ***[M2 2015 Q.12]***

(a) In the figure, the curve Γ consists of the curve *AB*, the line segments *BC* and *CO*, where *O* is the origin, *B* lies in the first quadrant and *C* lies on the *x*-axis. The equ. of *AB* and *BC* are $x^2 - 4y + 8 = 0$ and $3x + y - 9 = 0$ respectively.



(i) Find the coordinates of *B*.

(ii) Let *h* be the *y*-coordinate of *A*, where $h > 3$. A cup is formed by revolving Γ about the *y*-axis. Prove that the capacity of the cup is $\pi(2h^2 - 8h + 25)$. ***(7 marks)***

(b) A cup described in (a)(ii) is placed on a horizontal table. The radii of the base and the lip of the cup are 3 cm and 6 cm respectively.

(i) Find the capacity of the cup.

(ii) Water is poured into the cup at a constant rate of 24π cm^3/s. Find the rate of change of the depth of water when the volume of water in the cup is 35π cm^3. ***(6 marks)***

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HKCEE Questions (level 2)

7. ***[A.Math 1999 Paper II Q.13]***

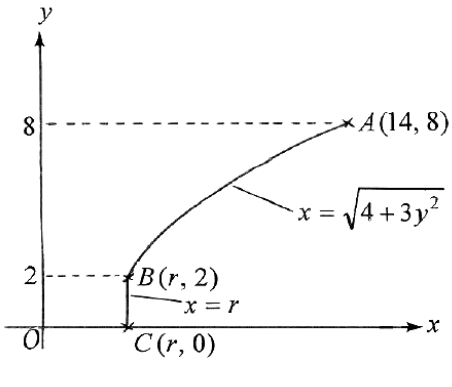


Figure 6(a)

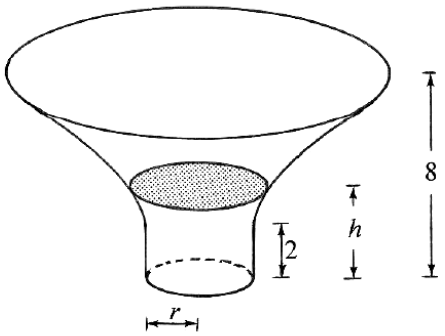


Figure 6(b)

A curve passes through three points $A(14, 8)$, $B(r, 2)$ and $C(r, 0)$ as shown in Figure 6(a). The curve consists of two parts. The equation of the part joining A and B is $x = \sqrt{4+3y^2}$ and the part joining B and C is the vertical line $x = r$.

(a) Find the value of r. ***(2 marks)***

(b) A pot, 8 units in height, is formed by revolving the curve and the line segment OC about the y-axis, where O is the origin. (See Figure 6(b).) If the pot contains water to a depth of h units, where $h > 2$, show that the volume of water V in the pot is $(h^3 + 4h + 16)\pi$ cubic units. ***(7 marks)***

(c) Initially, the pot in (b) contains water to a depth greater than 3 units. The water is now pumped out at a constant rate of 2π cubic units per second. Find the rate of change of the depth of the water in the pot with respect to time when

(i) the depth of the water is 3 units, and

(ii) the depth of the water is 1 unit. ***(7 marks)***

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HKCEE Questions (level 2)

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HKCEE Questions (level 2)

8. ***[A.Math 2006 Q.16]***

Let C be the curve $y = \frac{1}{3}x^2 - \frac{4}{3}x + 1$. C_1 is a part of C with $0 \le x \le 1$ and C_2 is a part of C with $3 \le x \le 4$.

(a) (i) Show that the equation of C_1 is $x = 2 - \sqrt{3y+1}$.

(ii) Write down the equation of C_2 in the form $x = f(y)$. ***(2 marks)***

(b)

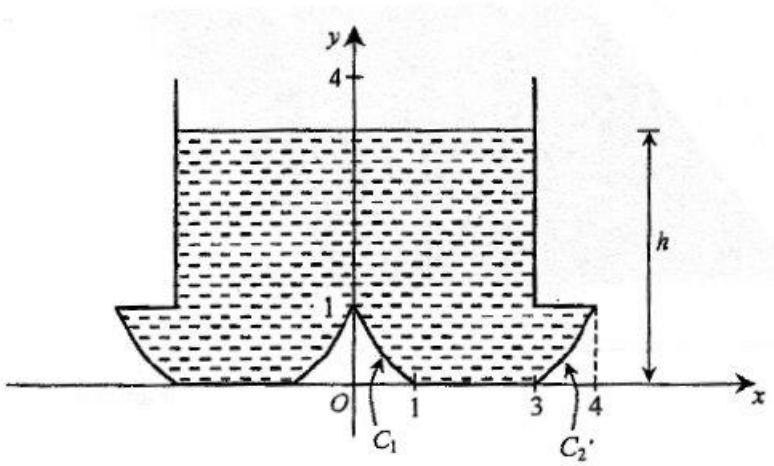


Figure 6

A container is formed by revolving C_1, the line segment $y = 0$ (for $1 \le x \le 3$), C_2, the line segment $y = 1$ (for $3 \le x \le 4$) and the line segment $x = 3$ (for $1 \le y \le 4$) about the y-axis (see Figure 6). Starting from time $t = 0$, water is poured into the container at a constant rate of 8π cubic units per minute. Let the volume and depth of water in the container at time t minutes be V cubic units and h units respectively.

(i) Consider $0 < h < 1$.

(1) Show that $V = \frac{16\pi}{9}[(3h+1)^{\frac{3}{2}} - 1]$.

(2) Find $\frac{dh}{dt}$ in terms of h.

(ii) Consider $1 < h < 4$. Find $\frac{dh}{dt}$.

(iii) It is known that $h = 1$ at $t = t_1$ and $h = 4$ at $t = t_2$. Sketch a graph to show how h varies with t for $0 \le t \le t_2$.
(You are not required to find the values of t_1 and t_2.) ***(10 marks)***

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HKCEE Questions (level 2)

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HKCEE Questions (level 3)

9. ***[A.Math 1996 Paper II Q.11]***

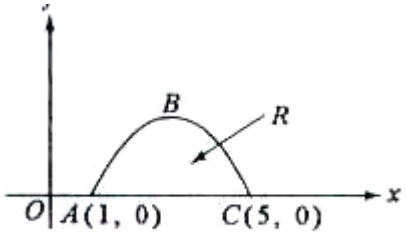


Figure 4 (a)

In figure 4(a), the region *R* is enclosed by the parabola $y = 4-(x-3)^2$ and the line segment *AC*, where *A* and *C* are the points (1,0) and (5,0) respectively. *B* is the vertex of the parabola.

(a) Write down the coordinates of *B*. ***(1 marks)***

(b) Show that the equation of the curve *AB* is $x = 3-\sqrt{4-y}$. ***(2 marks)***

(c)

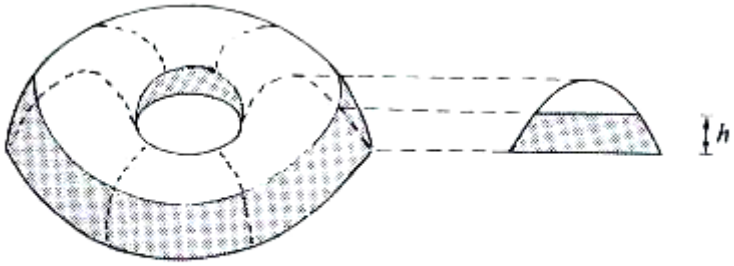


Figure 4(b)

A jelly ring is in the shape of the solid of revolution of the region *R* in Figure 4(a) about the *y*-axis. Furthermore, the jelly ring contains two layers .Let *h* be the height of the lower layer. (See Figure 4(b))

(i) Show that the volume of the lower layer of the jelly ring is $8\pi[8-(4-h)^{\frac{3}{2}}]$.

(ii) If the two layers have equal volumes, find the value of *h* correct to 3 significant figures. ***(9 marks)***

(d)

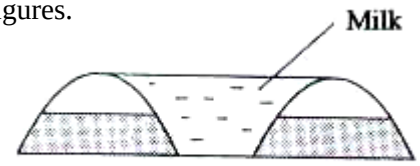


Figure 4(c)

If milk is poured into the center of the jelly ring in (c) until it is completely filled (see Figure 4(c)), find the volume of milk required. ***(4 marks)***

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HKCEE Questions (level 3)

10. ***[A.Math 2002 Q.16]***

(a)

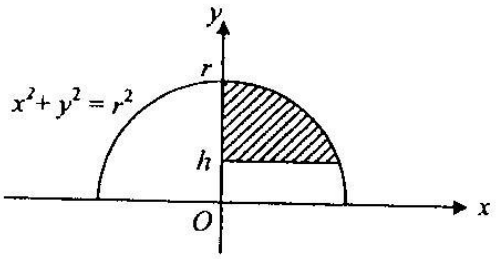


Figure 8

In Figure 8, the shaded region is bounded by the circle $x^2 + y^2 = r^2$, the *y*-axis and the line $y = h$, where $0 \le h < r$. The shaded region is revolved about the *y*-axis. Show that the volume of the solid generated is $\frac{\pi}{3}(2r^3 + 3r^2h + h^3)$. ***(4 marks)***

(b)

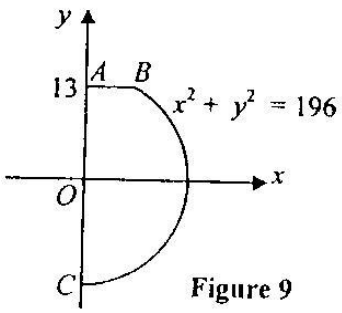


Figure 9

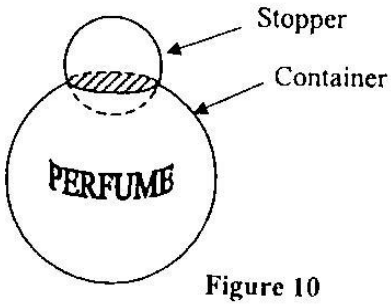


Figure 10

In Figure 9, *A* and *C* are points on the *y*-axis, *BC* is an arc of the circle $x^2 + y^2 = 196$ and *AB* is a segment of the line $y = 13$. A pot is formed by revolving *BC* about the *y*-axis.

(i) Find the capacity of the pot.

(ii) Figure 10 shows a perfume bottle. The container is in the shape of the pot described above and the stopper is a solid sphere of radius 6. Find the capacity of the perfume bottle. ***(8 marks)***

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HKCEE Questions (level 3)

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HKCEE Questions (Level 3)

11. ***[A.Math 1994 Paper II Q.13]***

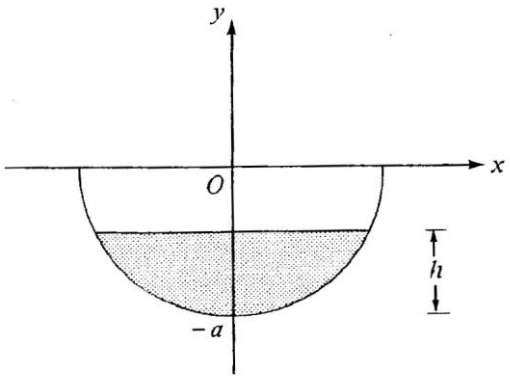


Figure 4

A bowl is generated by revolving the lower half of the circle $x^2 + y^2 = a^2$ about the *y*-axis. At tine *t*, the bowl contains water to a depth of *h*, where $0 \le h < a$ (see Figure 4). Let *V* be the volume of water in the bowl and *A* be the area of the water surface.

(a) Show that

(i) $V = \pi h^2 (a - \frac{1}{3}h)$,

(ii) $A = \pi h(2a - h)$. ***(7 marks)***

(b) Suppose the water evaporates at a rate proportional to the area of the water surface at that instant, i.e. $\frac{dV}{dt} = -kA$, where *k* is a positive constant.

(i) Show that $\frac{dh}{dt}$ is a constant.

(ii) It is given that the depth of the water is $\frac{3}{4}a$ at *t* = 0 and the water will completely evaporate at *t* = 30.

(1) Show that, for $0 \le t \le 30$, $h = \frac{a}{40}(30 - t)$.

(2) Find the volume of water in the bowl at *t* = 10. ***(9 marks)***

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HKCEE Questions (level 3)

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HKCEE Questions (level 3)

12. ***[A.Math 2003 Q.19]***

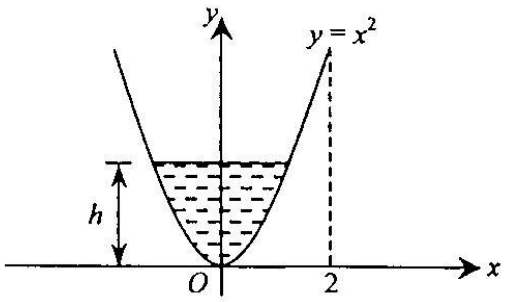


Figure 6

A water tank is formed by revolving the curve $y = x^2$, where $0 \le x \le 2$, about the *y*-axis (see Figure 6). Starting from time *t* = 0, water is pumped into the tank at a constant rate of 2π cubic units per minute. Let the volume and the depth of water (measured from the lowest point of the tank) in the tank at time *t* (in minutes) be *V* cubic units and *h* units respectively.

(a) Express *V* in terms of *h*.
Hence show that it takes 4 minutes to fill up the tank. ***(5 marks)***

(b) Show that $\frac{dh}{dt} = \frac{2}{h}$. ***(3 marks)***

(c) Which of the sketches in Figure 7 best describes the relation between *h* and *t*?
Explain your answer. ***(2 marks)***

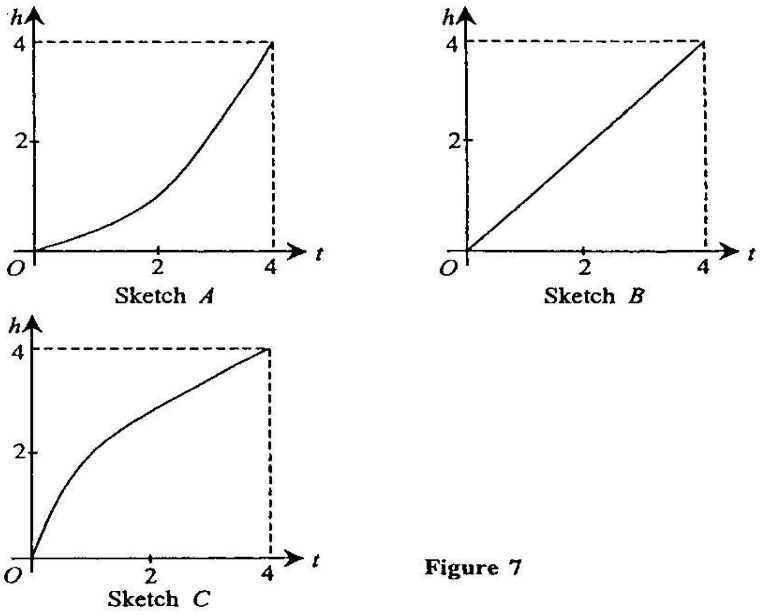


Figure 7

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HKCEE Questions (level 3)

12. ***[A.Math 2003 Q.19] (Cont'd)***

(d) An engineer decides to modify the tank by laying cement on the upper part of its interior wall, so that the interior of the tank becomes cylindrical in shape for $y \geq 2$ as shown in Figure 8. Water is pumped into the empty new tank at a constant rate of 2π cubic units per minute until it is full. Sketch the graph of h against t for the new tank in the same sketch you chose in (c). ***(2 marks)***

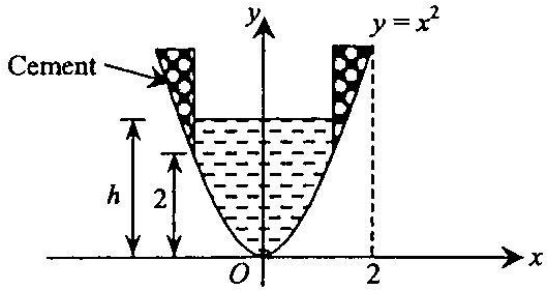


Figure 8

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13. ***[A.Math 2009 Q.15]***

(a)

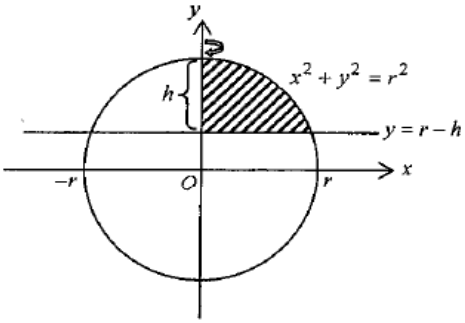


Figure 5

In Figure 5, the shaded region is bounded by the circle $x^2 + y^2 = r^2$, the *y*-axis and the straight line $y = r - h$, where $0 \le h \le 2r$. Show that the volume of the solid generated by revolving the shaded region about the *y*-axis is $\pi r h^2 - \frac{\pi h^3}{3}$.

(4 marks)

(b)

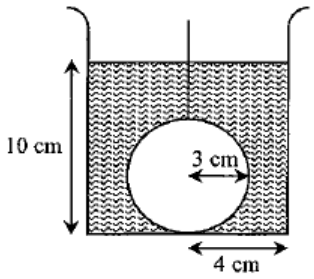


Figure 6

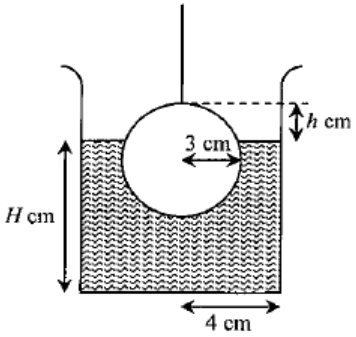


Figure 7

A metal sphere of radius 3 cm, with a thin string attached, is placed inside a circular cylindrical container of base radius 4 cm. Water is poured into the container until the depth of the water is 10 cm (see Figure 6). The sphere is then being pulled vertically out of the water. Let *H* cm and *h* cm be the depth of the water and the distance between the top of the sphere and the water surface respectively (see Figure 7).

(i) Prove that $H = \frac{1}{48}(h^3 - 9h^2 + 480)$.

(ii) The sphere is being pulled at a constant speed of $\frac{1}{4}$ cms^{-1}. At the instant when $h = 3$, find the rate of change of

(1) the depth of the water,

(2) the distance between the top of the sphere and the water surface.

(8 marks)

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HKCEE Questions (level 3)

14. ***[A.Math 2010 Q.16]***

(a)

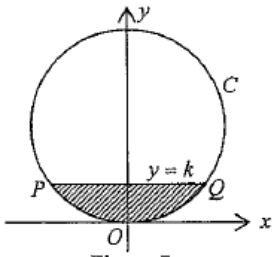


Figure 7

Figure 7 shows the shaded region bounded by the circle $C: x^2 + y^2 - 2ay = 0$ and the line $PQ: y = k$, where $0 \le k \le 2a$ and $a > 0$. Show that the volume of the solid by revolving the shaded region about the *y*-axis is $\pi\left(ak^2 - \frac{k^3}{3}\right)$.

(b) ***(3 marks)***

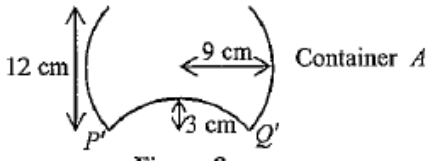


Figure 8

Figure 8 shows a container *A* in the shape of part of a sphere of radius 9 cm with the bottom part reflected about the horizontal plane passing through *P*' and *Q*'. The height of the container is 12 cm and the height of the bottom part of the container is 3 cm.

(i) Find the capacity of container *A*.

(ii)

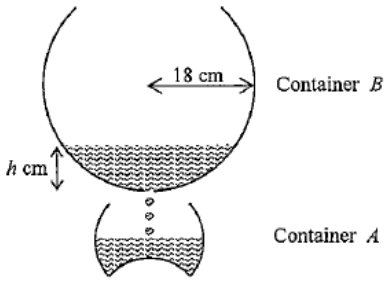


Figure 9

In Figure 9, container *B* is in the shape of part of the sphere of radius 18 cm. Initially, container *A* is empty and container *B* contains water of volume 909π cm^3. Water is leaking through a hole at the bottom of container *B* into container *A*. Let *h* cm be the depth of water in container *B*. ***(9 marks)***

HKCEE Questions (level 3)

14. ***[A.Math 2010 Q.19] (Cont'd)***

Consider the moment when container *A* is just full.

(1) Show that $h^3 - 54h^2 + 459 = 0$.

(2) If the volume of water in container *A* is increasing at rate 45π cms^{-1}, find the rate of decrease of water level in container *B*.

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HKCEE Questions (Level 3)

15. ***[A.Math 2007 Q.18]***

(a) It is given that the curve $y = 2\sqrt{x} - x$ has a horizontal tangent at $x = r$.
Show that $r = 1$.

(2 marks)

(b)

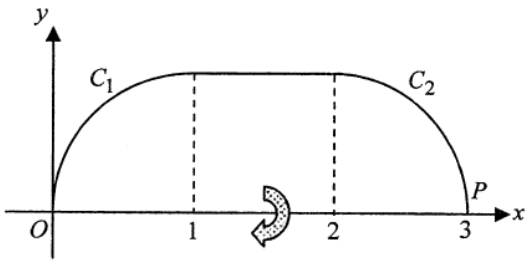


Figure 11

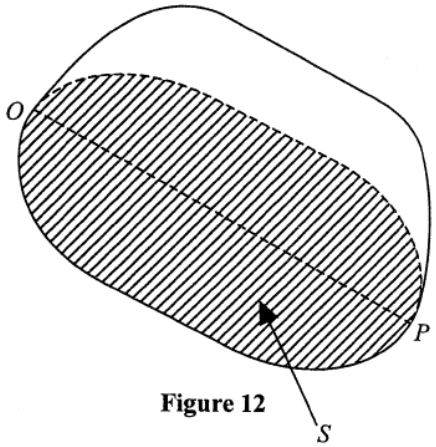


Figure 12

Let O be the origin and P be the point $(3,0)$. Figure 11 shows a region bounded by:

[1] the curve $C_1 : y = 2\sqrt{x} - x$ (for $0 \le x \le 1$),
[2] the line segment $y = 1$ (for $1 \le x \le 2$),
[3] the curve $C_2 : y = 2\sqrt{3-x} - (3-x)$ (for $2 \le x \le 3$), and
[4] OP.

Figure 12 shows a solid formed by revolving the region about the x-axis by $\mathbf{180°}$.

(i) The base of the solid is denoted by S in Figure 12. Find the area of S.

(ii) Show that the volume of the solid is $\frac{37}{30}\pi$.

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15. ***[A.Math 2007 Q.18] (Cont'd)***

(iii)

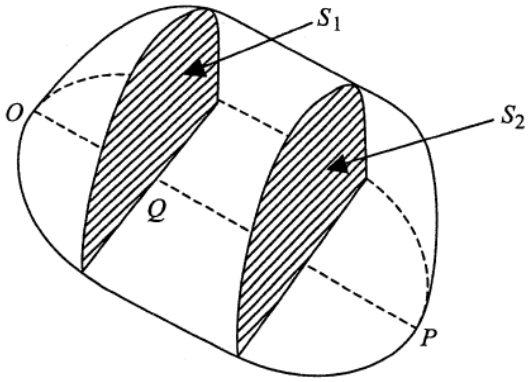


Figure 13

Mrs. Chan has baked a cake which is in the shape of the solid in Figure 12. She cuts the cake into three parts of equal volumes for her three children. The cross-sections formed, S_1 and S_2, are perpendicular to OP (see Figure 13).

Let the intersection of OP and S_1 be Q.

Find $OQ : OP$.

(10 marks)

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HKCEE Questions (level 3)

16. ***[A.Math 2000 Paper II Q.11]***

(a)

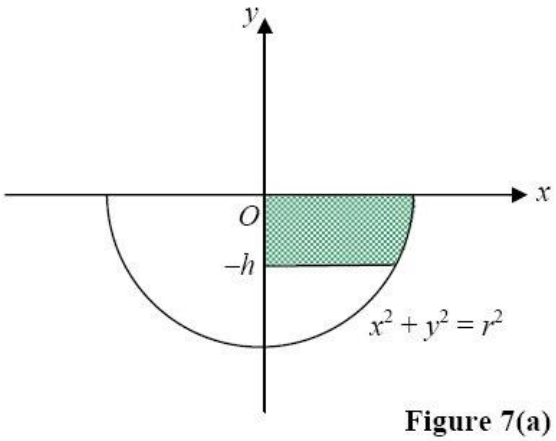


Figure 7(a)

In Figure 7(a), the shaded region is bounded by the circle $x^2 + y^2 = r^2$, the *x*-axis, the *y*-axis and the line $y = -h$, where $h > 0$. If the shaded region is revolved above the *y*-axis, show that the volume of the solid generated is $(r^2h - \frac{1}{3}h^3)\pi$ cubic units.

(4 marks)

(b)

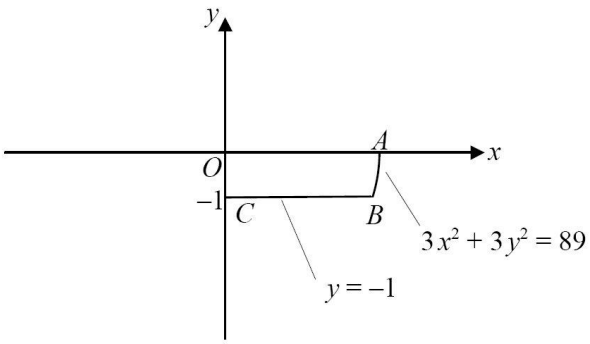


Figure 7(b)

In Figure 7(b), *A* and *C* are points on the x-axis and y-axis respectively. *AB* is an arc of the circle $3x^2 + 3y^2 = 89$ and *BC* is a segment of the line $y = -1$. A mould is formed by revolving *AB* and *BC* about the y-axis. Using (a), or otherwise, show that the capacity of the mould is $\frac{88}{3}\pi$ cubic units. ***(2 marks)***

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16. ***[A.Math 2000 Paper II Q.11] (Cont'd)***

(c)

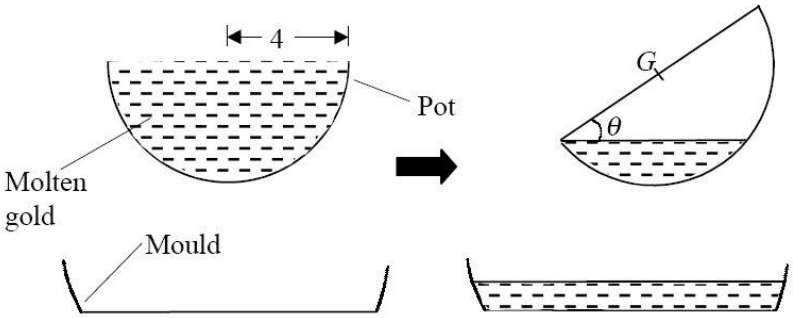


Figure 7(c)

Figure 7(d)

A hemispherical pot of inner radius 4 units is completely filled with molten gold. (See Figure 7(c).) The molten gold is then poured into the mould mentioned in (b) by steadily tilting the pot. Suppose the pot is tilted through an angle θ and G is the center of the rim of the pot. (See Figure 7(d).)

(i) Find, in terms of θ,

(1) distance between G and surface of the molten gold remaining in pot,

(2) the volume of gold poured into the mould.

(ii) When the mould is completely filled with molten gold, show that

$$8\sin^3\theta - 24\sin\theta + 11 = 0.$$

Hence find the value of θ. ***(10 marks)***

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